

The empirical recurrence for OEIS A254163, recovered from its tail

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Abstract

OEIS A254163 records “Empirical recurrence of order 66” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 464$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 6$, so the recurrence is proved for every $n \geq 72$. Its last coefficient is $c_{66} = 4294967296 \neq 0$, so no further tail satisfies anything shorter; any order-66 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A254163 is “Number of $(n+2) \times (3+2)$ 0..1 arrays with every 3×3 subblock diagonal maximum minus antidiagonal median nondecreasing horizontally, vertically and ne-to-sw antidiagonally.”. It has offset 1 and begins

16240, 164408, 1553720, 13514192, 119471400, 1054310240, 9378850720, 83642043104, ...

The entry states:

Empirical recurrence of order 66 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 14:28:57, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 464$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 464$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 66: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 6$ is the first whose minimal order is exactly the stated 66.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 928$ exact terms from $n = 6$ on, it returns order 66 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{66} c_i a(n-i),$$

c_1	10	c_{18}	99502736	c_{35}	1996849152	c_{52}	2013265920
c_2	1	c_{19}	−44703040	c_{36}	−2664361984	c_{53}	6006243328
c_3	−104	c_{20}	−141757056	c_{37}	2099658752	c_{54}	−6392119296
c_4	121	c_{21}	210736960	c_{38}	2366996480	c_{55}	−6845104128
c_5	−210	c_{22}	147969152	c_{39}	−2840068096	c_{56}	−6442450944
c_6	−345	c_{23}	−1016721664	c_{40}	−4047503360	c_{57}	536870912
c_7	5012	c_{24}	699197568	c_{41}	−7393509376	c_{58}	5905580032
c_8	10272	c_{25}	269498368	c_{42}	11170807808	c_{59}	6442450944
c_9	−148308	c_{26}	−512012544	c_{43}	−4787535872	c_{60}	−6710886400
c_{10}	124984	c_{27}	−398208512	c_{44}	−2981625856	c_{61}	4294967296
c_{11}	157552	c_{28}	2012269824	c_{45}	11083710464	c_{62}	0
c_{12}	−301448	c_{29}	924857856	c_{46}	4004511744	c_{63}	−5368709120
c_{13}	1112064	c_{30}	−3409479936	c_{47}	−12050235392	c_{64}	−4294967296
c_{14}	−8826160	c_{31}	4126692352	c_{48}	4697620480	c_{65}	0
c_{15}	80887520	c_{32}	407853056	c_{49}	−11765022720	c_{66}	4294967296
c_{16}	−67956944	c_{33}	−2900448256	c_{50}	1434451968		
c_{17}	−83228256	c_{34}	−1580879872	c_{51}	17418944512		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 72$, and it is the recurrence OEIS A254163 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 6$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{66} - c_1 x^{65} - \dots - c_{66}$. Its constant term is $-c_{66} = -4294967296$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the

components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 66 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 66, so $\deg q = 66$, and it is monic. It holds from some index t on. From $\max(t, 6)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 66, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 15 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 66, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 4294967296.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-66 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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